

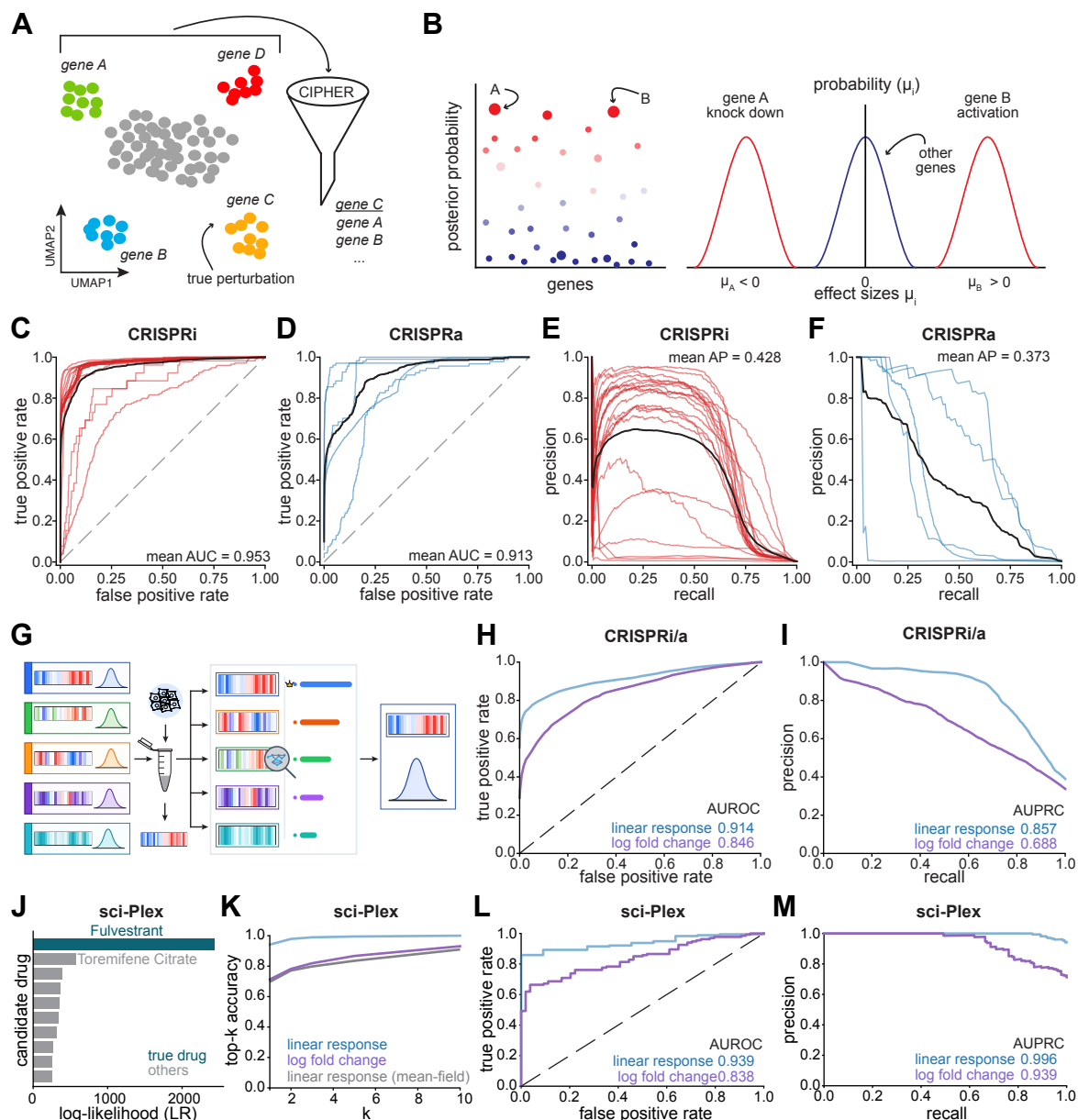


Fig. 5: The inverse problem: predicting causal drivers of perturbation response in transcriptome-wide screens.

a, Schematic depicting how CIPHER ranks genes responsible for a measured perturbation using knowledge of the control-cell gene expression fluctuations. Each point corresponds to a transcriptome from a single cell and is colored by the perturbed gene. **b**, Schematic of example data and interpretation. Left: Posterior probability of a nonzero perturbation effect across genes (points). True perturbed genes A and B have high posterior probabilities. Right: Effect size distributions for different perturbations. Distributions of effect size for genes A and B as well as a composite distribution over other unperturbed genes. **c**, Receiver-operator characteristic curves for each CRISPRi dataset (colored curves, mean AUC = 0.953). Black curve indicates average across datasets. **d**, Receiver-operator characteristic curves for each CRISPRa dataset (colored curves, mean AUC = 0.913). Black curve indicates average across datasets. **e**, Precision-recall curves for each CRISPRi dataset (colored curves, mean AP = 0.428). Black curve indicates average across datasets. **f**, Precision-recall curves for each CRISPRa dataset (colored curves, mean AP = 0.373). Black curve indicates average across datasets. **g**, Schematic of the likelihood-based CIPHER inverse ranking of transcriptional perturbation signatures. CIPHER identifies that the true signature is blue. **h**, Receiver-operator characteristic curves composite for all perturbations across CRISPR datasets for CIPHER (blue, linear response; 0.914) and log-fold-change ranking (purple, 0.846). **i**, Precision-recall curves for CIPHER (blue, 0.857) and log-fold-change ranking (purple, 0.688) across CRISPR datasets. **j**, CIPHER log-likelihood ranking for one held-out drug in the sci-Plex⁶¹ experiment. True compound is shown in blue. **k**, Top-k accuracy for identifying the true sci-Plex⁶¹ compound from heldout cells with CIPHER (blue) and log-fold-change ranking (purple). Mean-field linear response is shown in gray. **l**, Receiver-operator characteristic curves composite for all perturbations across sci-Plex⁶¹ for CIPHER (blue, 0.939) and log-fold-change ranking (purple, 0.838). **m**, Precision-recall curves for CIPHER (blue, 0.996) and log-fold-change ranking (purple, 0.939) across sci-Plex⁶¹ perturbations.